

Seeking Principled Diffeomorphic Hylogisms (4/6): Symmetries, Discretisation, and Lattice Formation in Conventional Semantics Recovery

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The prior iteration contained several errors: (i) the NWET and h-principle were conjectured necessary for semantic disentanglement but are succeeded by a simpler triangle inequality/non-degeneracy explanation, (ii) spectral decomposition claims have yielded to a more comprehensive discussion of discretisation techniques. The author's approach to this 6-part series is to post each part's technical sketch first without proofs, presuming at each step that prior sections hold. Should later scrutiny invalidate prior sections, the author will diminish or amend claims accordingly. The reader should not treat these early iterations as settled.

Abstract

This research note is the fourth in a series that examines the bridge between the physical universe's symmetric informational descriptions. En-route to a diffeomorphic connection from the state-configuration space to the semantic vector space, this note seeks a functorial hylogism, allowing the principled identification of physical structures across space and scales. The prior note organised the s-catalogue into comprehensive vector spaces ($\mathbb{V}_{int.}, \mathbb{V}_{ext.}$) via semantic-warping interactions. This note discusses how conventional semantic spaces are retrieved under quotienting, and how discretisation yields principled naming conventions.

Semantic Warping and Meaning

Compression, Stretch, and Distance-Centroids

Scholars often posit that semantics lie in the compression of symmetric information across physical or informational elements [1], [2]. For example, 'chair' refers to a broad set of chair-like physical objects¹, and now their vector encodings in $\mathbb{V}_{int.}$ and $\mathbb{V}_{ext.}$. So, one might conclude that the label 'chair' somehow compresses these vectorised instances, and even that this compression *must occur* for a set of semantic labels to carry meaning. A label's meaning becomes the objects to which the label refers.

Notice, though, that external labelling is at odds with the current formalism. In $\mathbb{V}_{int.}$, each label is a physical object itself. Person A's 'chair' differs from B's 'chair'; the words are uttered at different times and from different mouths, each oscillating different air particles; a label is not a single abstract signifier. 'So', one asks, 'how can these different labels mean the same thing to an observer?' Firstly, the semantic warping tensor determine's the observer's Riemannian metric g_{obs} on $\mathbb{V}_{ext., obs.}$. For two physical forms to bear semantic similarity, the forms must reside within a close semantic neighbourhood, dictated by obsever-interactions

$$d_{obs}(v_1, v_2) = \inf_{\gamma} \int_0^1 \sqrt{g_{obs}(\dot{\gamma}(t), \dot{\gamma}(t))} dt < \epsilon \quad (1)$$

¹(e.g., from armchairs to rocking chairs)

The intrinsic semantic vector space therefore has no blanket labels like ‘chair’; only ‘chair’ utterances and chair-like objects, each label or object distinguished locally from elements of the same class by infinitesimal deviations in physical form. Then, these symmetric ‘chair’ labels are compressed by the observer’s warping tensor such that their metric-adjusted distances in $\mathbb{V}_{ext,obs}$ reduce. Objects that bear little physical similarity but have essential semantic similarities suddenly occupy local neighbourhoods. Conversely, objects with minute physical differences² but extreme semantic differences diverge drastically to denote brutal semantic transitions.

From this view, a person may understand a ‘chair’ as all objects within a distance d from a semantic centroid v_{chair} ,

$$\mathcal{C}_{chair} = \{[s] \in \mathcal{E}_{\mathbb{S}} \mid d_{obs}(h([s]), v_{chair}) \leq d\} \quad (2)$$

where, for example, one locates the centroids via k-means or segments the space as a Voronoi tessellation.

Disentanglement Guarantees

‘Wait,’ the reader interjects, ‘how can one cluster all the different physical forms semantically when each may differentially relate to objects perceived as brutally different?’ Fortunately, the nature of the pullback metric ensures that objects may disentangle without tearing the space or creating singularities. To see how, note that changing the Riemannian metric on a fixed smooth space does not alter the space’s underlying topology or smooth structure. The metric g_{ext} on the extrinsic space³ satisfies the triangle inequality:

$$d_{ext}(h(v_1), h(v_2)) + d_{ext}(h(v_1), h(v_3)) \geq d_{ext}(h(v_2), h(v_3)) \quad (3)$$

$$\text{and then } d_{sem}(v_1, v_2) + d_{sem}(v_1, v_3) \geq d_{sem}(v_2, v_3). \quad (4)$$

The distance induced by the pullback metric $g_{sem} = h^*g_{ext}$ also satisfies the triangle inequality when the intrinsic metric warps $g_{int} \rightarrow g_{sem}$. One may go further to prove that deforming the intrinsic space with non-degenerate metrics protects $\mathbb{V}_{int}$ ’s topology.⁴⁵ One may expand and simultaneously compress metrics between distinct objects (anisotropic scaling) while retaining the local diffeomorphism $\mathbb{V}_{int,l} \leftrightarrow \mathbb{V}_{ext,l}$; physically similar forms disentangle along semantic directions without tearing the space $\mathbb{V}_{int,l}$.

Note, however, that anisotropic scaling only provably preserves topology at every finite-dimensional truncation for which g_{sem} remains positive-definite. In the infinite-dimensional limit, however, asymptotically vanishing metric weights may induce a strictly weaker topology; no general topological-preservation guarantee is presently claimed for that limit.

Geometric Partition Challenges

The issue with using g_{sem} to identify semantic regions is that the resultant discretisation remains unconstrained. Suppose, for example, that two compressed regions under g_{sem} share a basin bounding the direction of compression. This boundary contour seems a plausible separator for objects compressed either side. Yet, a metric alone does

²(e.g., a fir tree with a star placed atop, or a chair with minute hairline fractures spread across each leg)

³Whose hylogism image is the difference vectors of an observer relative to the un-probed baseline.

⁴Non-degeneracy was guaranteed in part 3 by assuming that dh_v is injective at every $v \in V_{int}$.

⁵

Proof. Since g_{int} and g_{sem} are smooth, positive-definite metrics on the same finite-dimensional space, they are locally equivalent. So, around every $v \in V_{int}$, there exist constants $0 < c_v \leq C_v < \infty$ such that

$$c_v g_{int}(u, u) \leq g_{sem}(u, u) \leq C_v g_{int}(u, u) \quad (5)$$

for every local tangent vector u . Consequently, their induced distances bound one another locally, such that each metric generates the same open neighbourhoods. Hence replacing g_{int} with g_{sem} preserves the topology of V_{int} . $\square$

not select a uniquely privileged categorisation: one requires additional choices on how to recover the compression scalar field and basin [3], [4], [5]. So, while not entirely arbitrary, the categorisation is still choice-dependent.

The earlier centroid approach [2] also lacks a canonical placement and assigns clusters under modelling assumptions. The discretionary nature of these partitions creates an unstable relation between consequent linguistics and their referent physical forms. Worse still, myopic geometry-only grouping fails to ensure grounding via observer impressions - a principle fault.

Principled Discretisations

Non-Identifiability

One may formalise the notion that categorisation techniques are not generally canonical. To do so, fix the hylogic pipeline h , and an assumption object Λ . The assumption-induced optimal discretiser $D_{h,\Lambda}^*$ and its information state X are:

$$X := (V_{\text{int}}, g_{\text{sem}}), \quad D_{h,\Lambda}^*(X) := C_{h,\Lambda}^* \in \text{Part}(V_{\text{int}}), \quad (6)$$

$$\kappa_{h,\Lambda}^* : V_{\text{int}} \longrightarrow I_{h,\Lambda}, \quad \kappa_{h,\Lambda}^*(v) = i \iff v \in C_i^*, \quad C_i^* \in C_{h,\Lambda}^*. \quad (7)$$

The assumption object is an invented constraint on the space of possible discretisers that chooses the ideal categorisation. The framework currently lacks a theoretically principled assumption that identifies the canonical discretiser. Without Λ , no technique – however mathematically appealing – can be said to identify a unique principled categorisation. With Λ , the set of techniques are different means of converging to the unique optimum. So, inflection loci, compression basins, Morse complexes, spectral partitions, and Voronoi tessellations are principled, only insofar as they provably recover or converge to $C_{h,\Lambda}^*$.

$\mathcal{C}$ is the set of all possible categories, whose cardinality one need not specify in advance. Only the realised outputs matter, so the target partition is the collection of non-empty fibres

$$C_{h,\Lambda}^* := \left\{ (\kappa_{h,\Lambda}^*)^{-1}(\{i\}) : i \in \text{im}(\kappa_{h,\Lambda}^*) \right\} \in \text{Part}(V_{\text{int}}). \quad (8)$$

In other words, the assumption object specifies the correct group, but names, orders, and indices are still unconstrained. A geometric discretisation method $\widehat{D}_\mu$ receives only X and returns a candidate partition

$$\widehat{C}_\mu(X) := \widehat{D}_\mu(X) \in \text{Part}(V_{\text{int}}). \quad (9)$$

where μ may be a construction based on the aforementioned discretisers.

Theorem 1 (No Assumption-Free Recovery). *Fix X , and let Λ_X be the collection of admissible assumption objects compatible with X . Suppose:*

$$\exists \Lambda_1, \Lambda_2 \in \Lambda_X \quad (10)$$

such that

$$C_{h,\Lambda_1}^* \neq C_{h,\Lambda_2}^*. \quad (11)$$

Then no purely X -dependent discretisation method guarantees recovery of the target partition $\forall \Lambda \in \Lambda_X$.

Proof. Because D_μ receives the same X under Λ_1 and Λ_2 , D_μ returns the same candidate partition $\hat{C}_\mu(X)$ in both cases. A single partition cannot equal two distinct target partitions. Hence D_μ must fail to recover the target for at least one assumption object (given $\Lambda_1 \neq \Lambda_2$). $\square$

The theorem does not imply that a geometric construction is internally ill-defined. Taking the metric flow case, once the scalar statistic, parameters, and basin rule are fixed, one may determine a unique geometric partition. Still unconstrained is the identification of that partition with $C_{h,\Lambda}^*$, rendering the reconstruction a candidate, rather than canon. The canonicity condition rendering a geometric construction principled becomes:

$$D_\mu(X) = C_{h,\Lambda}^*, \quad (12)$$

up to category cell relabelling/permutation. For an approximation family $D_{\mu,\eta}$, the requirement becomes

$$d_{\text{part}}(D_{\mu,\eta}(X), C_{h,\Lambda}^*) \longrightarrow 0 \quad \text{as } \eta \longrightarrow 0. \quad (13)$$

Conversely, if the theory offers a unique rule F such that

$$C_{h,\Lambda}^* = F(X) \quad (14)$$

for every assumption object compatible with X , then geometry alone *does* identify the target map. Establishing the rule is itself an additional substantive assumption or recovery theorem; one that does not follow magically from the existence of $\mathbb{V}_{int}$'s geometric structure⁶.

Assuming Observer-Relative Attractor Equivalence

Analogously to the assumption of relative causal meaning, this section imposes a new assumption for discretisation:

Assumption 1 (Observer-Relative Attractor Equivalence). *An observer $v_{obs} \in \mathbb{V}_{int}$ perceives two probe objects v_{p_1}, v_{p_2} as occupying a single semantic category C_i if the observer's physical response dynamics, relative to the probe-interactions, occupy a shared attractor $\{v_{obs,t}\}_\nu$ where $v_{obs,t} \in \mathbb{V}_{int}^{3D}$.⁷*

Attractor-equivalence can be shown in the semantic space $\mathbb{V}_{int}$ as well as the observer's internal cognitive phase space $\mathcal{H}_{obs}$; indeed, the inter-representational mapping between each must preserve the attractor-dynamics. In general, describe the dynamics with an explicit formalism. Assume the observer's internal state progresses according to a state-variable PDE:

$$\dot{v}_{obs} = f(t, v_{obs}; v_p) \quad (15)$$

An operator χ retrieves latent pre-discretisation structure from the dynamics in the infinite limit. Require that smooth probe-deformations cause smooth latent structure deformations:

$$\text{Requirement 1: } \chi(v_p) = \lim_{t \rightarrow \infty} (f), \quad \delta v_p \implies \delta \chi(v_p) \quad (16)$$

where critical points on the pre-constructor domain cause category transitions. In addition, require that discretisers acting on the latent structure have a coarsening parameter η :

$$\text{Requirement 2: } C_p^* = \mathcal{D}_\eta(\chi(v_p)). \quad (17)$$

Adjusting the smooth coarsener lets the discretiser quotient the space at varying degrees of granularity, producing ever more inclusive categories.

⁶ $\mathbb{V}_{int}$ does not naturally identify semantic regional partitions, and the current work has required two separate assumptions (of which theorem 1 describes the latter.) The first assumption of observer-relative causal meaning induces the *g_{semantic} warp*. To constrain the resultant geometry, a second assumption is needed that selects the correct categorisation over elements.

⁷ $\mathbb{V}_{int}^{3D}$ is the intrinsic SVS resulting from a base space whose dimensionality is restricted to the three spatial dimensions.

References

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- [6] Gemini 3.1 Pro-Extended and Pascal Tohouri, *[public research] principled discretisations via stable attractors*, <https://share.gemini.google/qJJ9p6Adnba7>, Accessed: 2026-08-17, 2026.

Appendices

A Acknowledgments and Declarations - AI Use

AI contributed to this work. Table 1 details those contributions:

Date	Model	Contribution
2026-08-17	G-3.1-P-E	Gemini drafted footnote 6 on two-tiered no assumption free recovery. The author reworded. Gemini corrected the current table format and the chat bibtex. See: [6].
2026-08-18	G-3.1-P-E	Gemini scrutinised drafted sections and proposed changes to the general formalism after mutual deliberation. The author re-drafted. See: [6]

Models used: G-3.1-P-E - Gemini 3.1 Pro Extended.

Table 1: AI contributions by date and model.